



Scientific priorities and shepherds' perceptions of ungulate's contributions to people in rewilding landscapes

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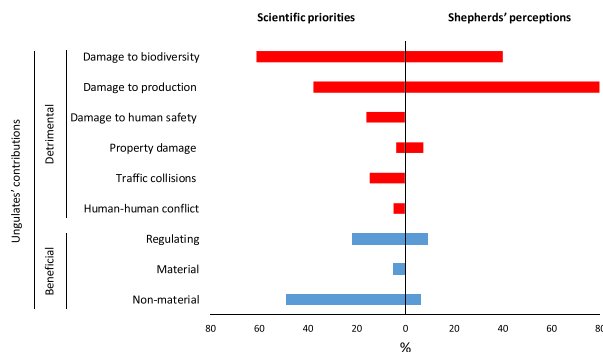
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HIGHLIGHTS

- Ungulates play a dual role as beneficial and detrimental contributions to people.
- Researchers and shepherds highlight detrimental aspects related with wild ungulates.
- However, both stakeholders stood out different ungulates' contributions to people.
- These results could affect the social tolerance towards ungulates.

GRAPHICAL ABSTRACT



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ABSTRACT

Nature's contributions to people (NCP) are all the contributions of living nature, both beneficial and detrimental, to the societies' life's quality. Ungulates play this dual role of providers of beneficial and detrimental NCP, as they are responsible of the supply of benefits (e.g. extractive experiences, habitat maintenance) and detriments (e.g. crops damage, traffic collisions). Our aim was to analyze the NCP provided by wild ungulates through examining the scientific priorities and the shepherds' perceptions in peninsular Spain. We reviewed scientific articles of NCP provided by ungulates in Spain and conducted questionnaires regarding NCP to shepherds in farming systems where domestic and wild ungulates cohabit. Then, we compared whether the scientific priorities match with those perceived by shepherds. Both stakeholders highlight more detrimental than beneficial NCP, although there are some mismatches between scientific priorities and shepherds' perceptions. Regarding detrimental NCP, soil alteration, silvicultural damage, human safety or traffic collision were included in scientific literature but not mentioned by shepherds. Contrarywise, shepherds mainly considered grazing competence and damage to animals (i.e. game species and livestock) as important detrimental NCP. Concerning beneficial NCP, whilst hunting was prominent in the publications, shepherds did not conceived it as an important beneficial contribution and considered the regulation of organisms (i.e. scavenging alternative prey) important benefits. These results can have twofold implications. The emphasis on detrimental NCP (studied and perceived) can reinforce the idea that ungulates can threaten humans rather than contribute to societies' wellbeing. The fact that research does not address the interests of shepherds can affect the social tolerance towards ungulates as the damages

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experienced or perceived by shepherds are not studied. Our results show the relevance of considering local knowledge held by shepherds and their perceptions, something highlighted by the NCP approach.

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1. Introduction

In the last decades, it has been acknowledged that biodiversity and its conservation contribute to societies' wellbeing through the provision of multiple ecosystem services (Brooks et al., 2006; Hevia et al., 2017). Likewise, there is also growing recognition that biodiversity can also undermine human wellbeing by causing natural hazards and impairing damages in livelihoods (e.g. Lyytimäki, 2015; Shackleton et al., 2016). The fact that biodiversity can both positively and negatively affect human wellbeing has been recently acknowledged by the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services through the concept of nature's contributions to people (NCP), i.e., all the contributions of nature, both beneficial and detrimental, to the societies' quality of life (Díaz et al., 2018). Frequently, the outcome of an ecological process, species or functional guild is conceived by society as a benefit or as a detriment depending on the environmental and cultural contexts (Saunders and Luck, 2016; Rasmussen et al., 2017). For example, scavengers are perceived by shepherds as beneficial because the removal of carcasses and at the same time as harmful because some carnivore scavengers also attack livestock (Morales-Reyes et al., 2018).

Whilst scientific attention on beneficial NCP provided by wildlife to societies are scarcely garnered (Lyytimäki, 2015; Shackleton et al., 2016), there is a large body of literature about the damages caused by wildlife to human wellbeing under the framework of human-wildlife conflicts, i.e. detrimental NCP (Peterson et al., 2010). The framework of human-wildlife conflicts refers to wildlife damages to food production, human safety, human properties or more generally livelihoods (Nyhus, 2016). Despite the provision of both beneficial and detrimental NCP by wildlife, yet the dual role of wildlife as beneficial and detrimental to human wellbeing is underexplored in the scientific literature.

In addition, the research on NCP is unevenly distributed among taxonomic groups. For example, whilst the role of vegetation for providing beneficial NCP has been broadly studied, the role of vertebrates remains understudied (Hevia et al., 2017). By contrast, research on detrimental NCP is biased towards carnivores (Peterson et al., 2010; Kinsky et al., 2014). There are other taxonomic groups that are relevant as providers of beneficial and detrimental NCP, such as vertebrate scavengers (Morales-Reyes et al., 2018) or rodents (Rasmussen et al., 2017). Ungulates are also a group that plays this dual role of providers of beneficial and detrimental NCP. Ungulates are recognized as key ecosystem engineers and they play a critical role for regulating NCP, such as habitat maintenance (Sandom et al., 2014; Barbosa et al., 2019), nutrient cycling (Abbas et al., 2012), seed dispersal (Gill and Beardall, 2001) and scavenging (Mateo-Tomás et al., 2015; Morales-Reyes et al., 2018). They also provide material NCP since they are still source of food in many societies (Ripple et al., 2015) and non-material NCP since they are a key element for nature-based tourism and recreational hunting (Naidoo et al., 2016). Ungulates can also be the source of detrimental NCP since they cause many damages, such as crops damage (Irby et al., 1996; Dunkley and Cattet, 2003), damage to forest plantations (Gundersen et al., 2004), collisions with vehicles (Snow et al., 2015) and sharing disease with livestock (Gortázar et al., 2007).

Wild ungulates are increasing in distribution and abundance in developed countries (Apollonio et al., 2010), and therefore, human-ungulate interactions is expected to rise under this rewilding scenario (Navarro and Pereira, 2015). The current absence of knowledge about the NCP provided by ungulates can entail a blind spot for wildlife management and biodiversity conservation, which can make any human-

wildlife coexistence difficult. Hence, the study of NCP provided by ungulates is especially relevant to give insights for their management.

In this research, we examine the scientific priorities and shepherds' perceptions regarding the provision of NCP by wild ungulates. Particularly, we analyzed the detrimental and beneficial NCP provided by wild ungulates through examining the scientific priorities and the shepherds' perceptions in peninsular Spain. We selected the wild ungulates of peninsular Spain because their populations are increasing favoured by changes in land use (Acevedo et al., 2011) that lead to competition for resources with extensive livestock farming. We specifically aimed to: (1) identify which detrimental and beneficial NCP provided by wild ungulates were included in the scientific literature of Spain; (2) determine whether wild ungulates were perceived as source of detrimental NCP or providers of beneficial NCP by shepherds in extensive and semi-extensive livestock farming systems in mountainous areas of Spain; and (3) compare whether the scientific priorities regarding detrimental and beneficial NCP of wild ungulates matched with the priorities derived from shepherds' perceptions. Furthermore, under a rewilding scenario we hypothesize that the distribution and ecological traits (diet, native/exotic, potential prey) of each ungulate species might influence both the scientific priorities and shepherds' perceptions.

2. Material and methods

2.1. Scientific priorities: systematic review of articles

We reviewed scientific articles of human-ungulates relations in Spain by considering beneficial NCP, detrimental NCP and a neutral relation between humans and ungulates. We searched in the Scopus database by using a search string comprised by words that combined different terms related to 'conflict' (i.e. detrimental NCP), 'ecosystem service' (i.e. beneficial NCP), 'human-herbivore relation' (i.e. neutral conceptualization of human-ungulates relations) and 'ungulate' (see Appendix A.1 in supplementary material for the full search string). The search was made to title, abstract and keywords from 2000 to 2018. We found 406 articles of potential relevance published in English. Of these, we only selected those articles of wild ungulates that are present in peninsular Spain (Fig. 1), excluding feral domestic ungulates. Eight species comprise the community of wild ungulates in peninsular Spain: the aoudad (*Ammotragus lervia*), the chamois (*Rupicapra pyrenaica*), the fallow deer (*Dama dama*), the Iberian ibex (*Capra pyrenaica*), the mouflon (*Ovis orientalis*), the roe deer (*Capreolus capreolus*), the red deer (*Cervus elaphus*), and the wild boar (*Sus scrofa*). We screened the articles to ensure that they reported on empirical studies (i.e., we excluded reviews or theoretical papers) and analyzed either detrimental or beneficial NCP provided by wild ungulates (i.e. whether the article studied or mentioned either detrimental NCP or beneficial NCP). This screening returned a final set of 82 articles (see Appendix A.2 in supplementary material).

We coded each publication according to: (1) its general description (i.e., year of publication, journal and the ungulate species studied), (2) category of detrimental NCP and (3) category of beneficial NCP. The classification of detrimental NCP was adapted from Peterson et al., (2010), but we included new detrimental NCP that were derived from the literature review (Table 1). We classified beneficial NCP according to the generalizing perspective provided by IPBES (Díaz et al., 2018; Table 2).

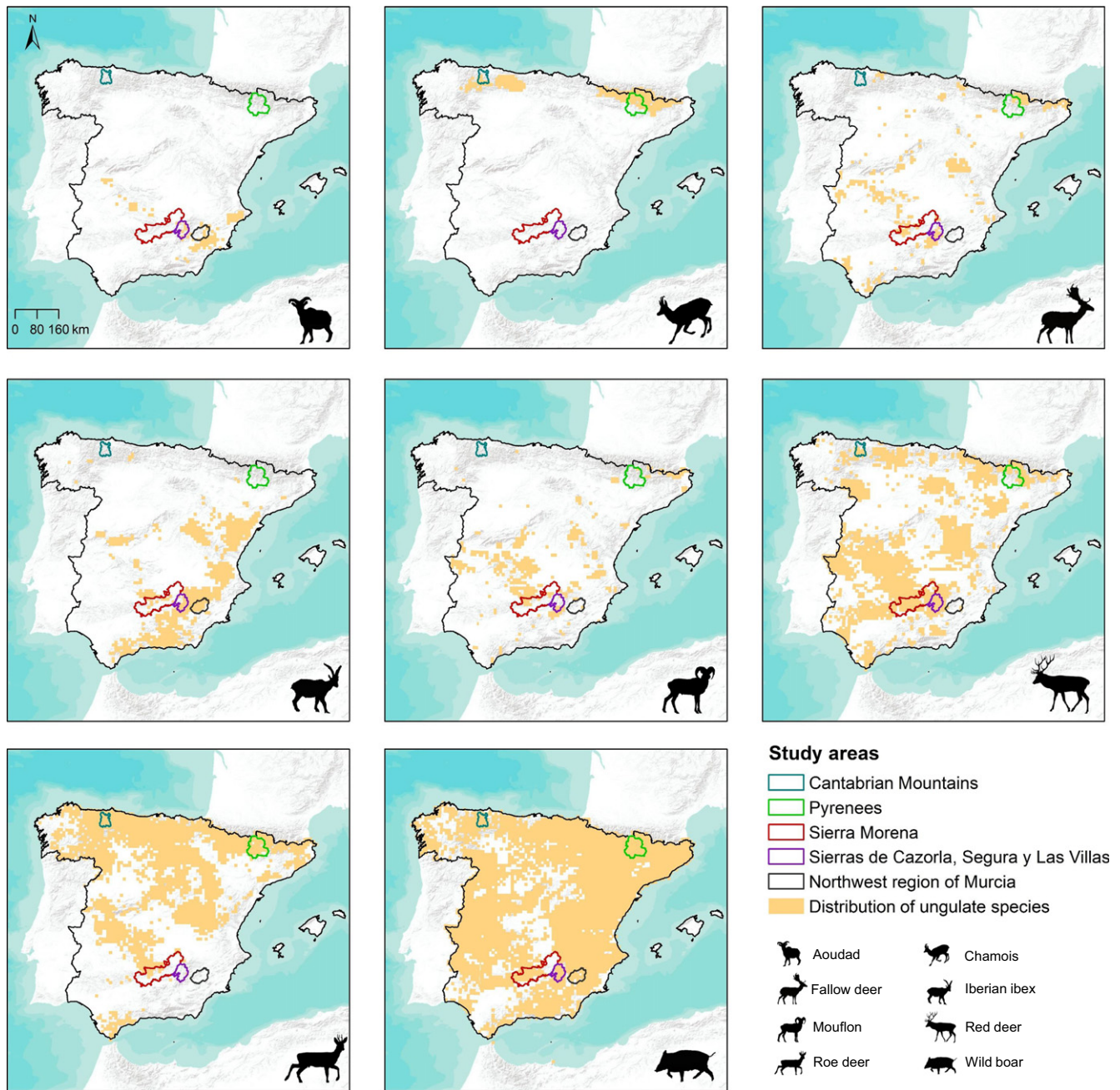


Fig. 1. Distribution map of each ungulate species in Spain.

2.2. Shepherds' perceptions

2.2.1. Study areas

We examined shepherds' perceptions regarding detrimental and beneficial NCP in five mountains in Spain (Fig. 1): the Cantabrian Mountains, the Pyrenees, Sierra Morena, Sierras de Cazorla, Segura y Las Villas Natural Park (hereafter, Cazorla) and Northwest region of Murcia (hereafter, Murcia). These areas represent the main traditional and large extensive and semi-extensive livestock farming systems (i.e., pasture-based farming) in Spanish mountains. These areas host an important number of livestock heads (Table A.1) that cohabit with wild ungulates.

Of the eight species comprising the community of wild ungulates in peninsular Spain, only the red deer and the wild boar are present in the five areas. The aoudad is a non-native species introduced into south-

eastern Spain in the 1970s and inhabits mountains of Cazorla and Murcia. The roe deer is present in the Cantabrian Mountains, Pyrenees and Sierra Morena. The Iberian ibex is present in the study areas of southern Spain (Sierra Morena, Cazorla and Murcia). The chamois is only present in the north of peninsular Spain (i.e. Cantabrian Mountains and Pyrenees). The fallow deer, probably another non-native species in peninsular Spain (see Baker et al., 2017), inhabits Cantabrian Mountains, Sierra Morena and Cazorla. The mouflon, also a non-native species introduced into Spain in the 1950s, occupies Pyrenees, Sierra Morena, Cazorla and Murcia (see Fig. 1 for detailed species distribution and Table A.2 for species considered in each study area). The distribution of wild ungulate species were extracted from the Spanish Inventory on Natural Heritage and Biodiversity (MITECO, 2013). Of the five study areas, stable populations of large carnivores, such as the brown bear (*Ursus arctos*) and the

Table 1
Categories of detrimental NCP related to wild ungulates in peninsular Spain extracted from the literature review and their consideration by the surveyed shepherds. The classification of detrimental NCP was adapted from Peterson et al., (2010).

Categories of detrimental NCP		Examples from scientific literature	Examples from shepherds
Damage to biodiversity	Vegetation damage	Negative effects on vegetation, including rooting (i.e. foraging activity within surface layers of soil) (Bueno and Jiménez, 2014).	Yes. "They root meadows", "They destroy the ground"
	Animal biodiversity damage	Negative effects on wild animal species without direct economic interest, such as lagomorphs (Carpio et al., 2014) or insects and arachnids (see Bernes et al., 2018).	Yes. "They displace other wild ungulates"
	Non-native	Exotic species are potentially harmful for the ecosystems. For instance, species introduction is related with biodiversity loss (Wilcove et al., 1998) and alteration of ecosystem processes and community structure (Hejda et al., 2009; Raizada et al., 2008).	Yes. "It is not native"
	Soil alteration	For instance, negative effects of wild ungulates on soil properties (Pascual-Rico et al., 2018).	No
Damage to production	Grazing competence	Wild ungulates consume pasture and other natural resources that could be used by livestock. For example, the European bison (<i>Bison bonasus</i>) competes directly with livestock (Kuemmerle et al., 2011).	Yes. "They eat the livestock pastures", "They eat the pastures that we pay"
	Diseases to livestock	Risk of illness transmission from wild ungulates to livestock (e.g. Acevedo et al., 2014).	Yes. "They transmit diseases to livestock: mange, tuberculosis, ticks..."
	Silvicultural damage	Impairment of natural forests or plantations intended for forestry (see Charco et al., 2016).	No
	Crop damage	Direct physical impairment of croplands and orchards (e.g. Giménez-Anaya et al., 2016).	Yes. "Damage to agriculture", "Destroy the crops"
Damage to human safety	Damage to animals	Direct physical damage caused by ungulates to livestock and big and small game species. For example, the wild boar as main nest predator of the common pheasant (<i>Phasianus colchicus</i>) (Senserini and Santilli, 2016).	Yes. "It feeds on game species", "It destroys red-legged partridges nests", "It attacks the cattle", "It kills lambs"
		Ungulates causing injuries, frightening and/or transmitting infections agents to humans (see García-Bocanegra et al., 2016).	No
	Property damage	Ungulates damaging human properties, particularly buildings and/or physical structures, such as fences (see Duarte et al., 2015).	Yes. "They damage the fences", "They shoot down walls and fences"
Traffic collisions		Ungulates damaging vehicles by ungulate-vehicle collision and/or causing traffic accidents (see Colino-Arrabal et al., 2018).	No
Human-human conflict		Conflict related to human disagreements over management decisions of wild ungulates or derived from different opinions and interests by different stakeholders (e.g. Gerhardt et al., 2013).	No

Table 2
Categories of beneficial NCP provided by wild ungulates in peninsular Spain extracted from the literature review and their consideration as benefits by the surveyed shepherds. Classification of NCP based on Díaz et al. (2018).

Categories of beneficial NCP		Examples from scientific literature	Examples from shepherds
Regulating	Habitat maintenance	The formation and continued production, by ungulates, of ecological conditions necessary or favourable for organisms important to humans (Díaz et al., 2018). E.g. to contribute to maintain semi-open habitats (Vera, 2000) and nutrient cycling (Danell et al., 2006).	Yes, as grazing: "Cleans the forest", "Generates pasture biodiversity in inaccessible areas for livestock"
	Dispersal of seeds	Facilitation by ungulates of seed dispersion of species important to humans. For example, grass or small herbs species via the coat, hoof or feces (Gill and Beardall, 2001).	No
	Maintenance of soils	E.g. contributing nutrients to the soil (Asner et al., 2004) and favour soil communities (Bueno and Jiménez, 2014).	Yes. "Wild boars aerate the soil by opening up holes"
	Regulation of hazards and extreme events	E.g. herbivory can contribute to fire prevention (Velamazán et al., 2018).	No
Material	Regulation of organisms	Removal of animal carcasses by wild boars, i.e., acting as a scavenger (Morales-Reyes et al., 2018), reduction of attacks on livestock due to the presence of alternative prey (Sidorovich et al., 2003) and wild ungulate carcasses as important food resource for vertebrate scavengers (Mateo-Tomás et al., 2015).	Yes, provision of both beneficial NCP, the wild boar as scavenger and ungulates as alternative preys and carcasses "They are wolves food and they [wolves] do not come for livestock"
	Food	Production of food from wild ungulates, such as meat from red deer (Milner et al., 2006).	Yes, "Wild boar cold meat in autumn"
	Materials and assistance	Materials obtained directly from ungulates as fur (see MacMillan and Phillip, 2008)	No
	Supporting identities	Source of satisfaction derived from knowing that a particular ungulate exists in the present (e.g. Martínez-Jauregui et al., 2015).	Yes. "It's so majestic that you have to protect it", "If they exist it's because of something", "They play their role"
Non-material	Physical and psychological experiences	Extractive experiences: Provision by ungulates of opportunities for physically beneficial leisure activities that entail extraction from nature, such as recreational hunting (Gamborg et al., 2017).	No
		Non-extractive experiences: Provision by ungulates of beneficial opportunities related with being in close contact with nature, such as the aesthetic value derived from species (de Pinho et al., 2014).	Yes. "I like to see them", "Beautiful: the most pretty and wild"
	Maintenance of options	Conservation of species associated with future situations as extinction of other populations (Fernández-Olalla et al., 2016).	No

Iberian wolf (*Canis lupus signatus*), are only present in the Cantabrian Mountains (Chapron et al., 2014; Palazón, 2017).

2.2.2. Data collection

We conducted 184 face-to-face questionnaires to shepherds from 2012 to 2016 (see Fig. A.1 for sampling points, Table A.1 for the sample size in each study area and Table A.3 for a description of variables obtained from the questionnaires). We structured the questionnaire in three main sections: (1) wild ungulate species that were known and seen by shepherds in the study area; (2) perception of NCP provided by wild ungulates in the study area; and (3) socio-demographic and farming characteristics of shepherds. We only asked questions concerning section two when respondents knew and had seen the ungulate species existing in the study area (see Appendix A.3 and Fig. A.2 in supplementary material for details about the species known and seen by shepherds in each study area).

To analyze the shepherds' perceptions on ungulate species, we built the variable *Nature's Contributions to People Provider (NCPPI) index*. It is based on the score assigned by each surveyed shepherd for each species present in their extensive livestock farming systems (see Table A.2). *NCPPI index* indicates the average shepherd perceptions of wild ungulates as providers of NCP for each species using a five-point scale from very harmful (*NCPPI index* = 1) to very beneficial (*NCPPI index* = 5). Based on this, species with an assigned *NCPPI index* ranged from 1 to 2 were considered harmful and, thus, as source of detrimental NCP; whilst species with *NCPPI index* ranged from 4 to 5 were considered providers of beneficial NCP. We carried out a Kruskal-Wallis test and Tukey post-hoc test to determine differences in the *NCPPI index* perceived by shepherds between different species. We also performed a Mann-Whitney *U* test to explore differences in the *NCPPI index* due to different ecological traits: (1) the diet of ungulates (herbivore or omnivore), (2) their origin (native or non-native) and (3) the presence of large predators in the livestock farming system (ungulates as potential prey). See Table A.4 for details on the different ecological traits (i.e. diet and origin) of the different ungulate species and for information on the presence of stable populations of large carnivores in each study area.

2.3. Comparison between scientific priorities and shepherds' perceptions

We compared the scientific priorities (percentage of publications including each NCP) matched with the priorities perceived by shepherds in extensive livestock farming systems (percentage of shepherds indicating that wild ungulates were responsible of providing specific NCP).

2.4. Statistical analysis

We tested if the mean number of beneficial and detrimental NCP provided by each species was correlated with the distribution of the species by performing Spearman's correlation test ($\alpha = 0.05$). For the scientific priorities, we calculate the mean number of NCP as the number of NCP included (studied or mentioned) in each paper divided by the total number of papers for each species. We calculated the distribution of the species based on the percentage of 10×10 km grids occupied by each species in peninsular Spain based on the Spanish Inventory on Natural Heritage and Biodiversity (MITECO, 2013). For the shepherds' perceptions, the distribution of the ungulate species was calculated as the percentage of 10×10 km grids that were occupied by each species in each study area.

To test whether there was significant association between the species of wild ungulates in peninsular Spain and the NCP included in the scientific literature, we conducted Chi-square contingency tables and Fisher's exact test ($\alpha = 0.05$). We conducted the same test to determine whether some wild ungulate species were particularly perceived by shepherds as a source of a specific NCP.

To compare the scientific priorities and the priorities perceived by shepherds in extensive livestock farming systems, we performed two Spearman's correlation tests ($\alpha = 0.05$): one for detrimental and another for beneficial NCP.

3. Results

3.1. Scientific publications on NCP provided by wild ungulates

Research on detrimental and beneficial NCP provided by ungulates in Spain has irregularly increased since 2000 (Fig. A.3). A great number of publications addressed both kinds of NCP (45.1% of publications). Of the remaining publications, most articles addressed only detrimental NCP (36.6%), whilst 18.3% of publications focused only on beneficial NCP. Species that received higher scientific attention were wild boars (61.0%), red deer (48.8%), roe deer (24.4%), Iberian ibexes (22.0%), fallow deer (14.6%) and aoudads (12.2%). Other species studied, although in <10% of publications, were the mouflon (7.3%), and the chamois (7.3%; Fig. 2).

Detrimental NCP that received higher attention were damage to biodiversity (61.0% of publications, which 40.3% were studied and 20.7% were only mentioned in the articles), particularly vegetation damage (54.9%) and damage to animal biodiversity (20.7%). Damage to production was the second most important detrimental NCP included in the scientific literature (37.8% of publications, which 18.3% were studied and 19.5% were only mentioned in the articles), especially disease transmission to livestock (18.3%). Damage to human safety was included in 15.9% of publications, which 3.7% were studied and 12.3% were only mentioned in the articles followed by traffic collisions (14.6% of publications, which 3.6% were studied and 11.0% were only mentioned in the articles). Damage to human safety specifically included disease transmission (15.9%) and direct attacks (2.4%). Property damage was included in <5% of publications (3.7% of publications, which were always studied in the articles.). Human-human conflicts derived from human relations with ungulates were included in 4.9% of publications, which were only mentioned in the articles. Table 1 and Fig. 3A present the detrimental NCP provided by ungulates as found in scientific literature.

The beneficial NCP that received the highest scientific attention belonged to the category of non-material NCP (48.8% of publications, which 3.7% were studied and 45.1% were only mentioned in the articles), being hunting (i.e., extractive experience) the most frequent non-material NCP. However, only 4.9% of publications mentioned the aesthetic value of wild ungulates (i.e., non-extractive experiences), just 1.2% the cultural heritage (i.e., supporting identities) and 1.2% the maintenance of options for the future. Regulating NCP were included in 23.2% of publications, which 14.7% were studied and 8.5% were only

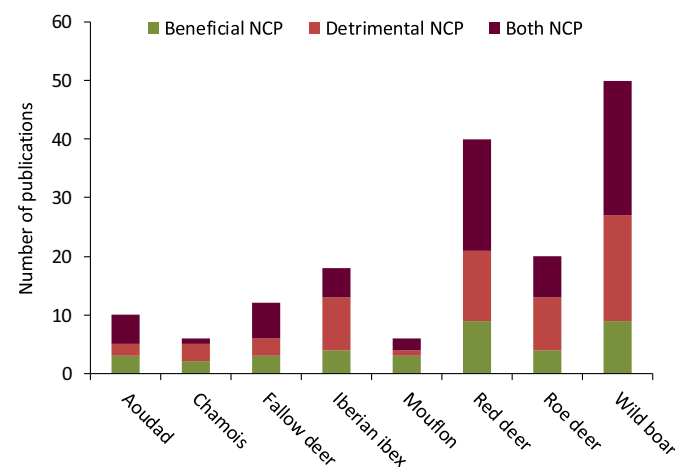


Fig. 2. Number of publications for each species. Bars indicate total number of papers focused on each NCP type.

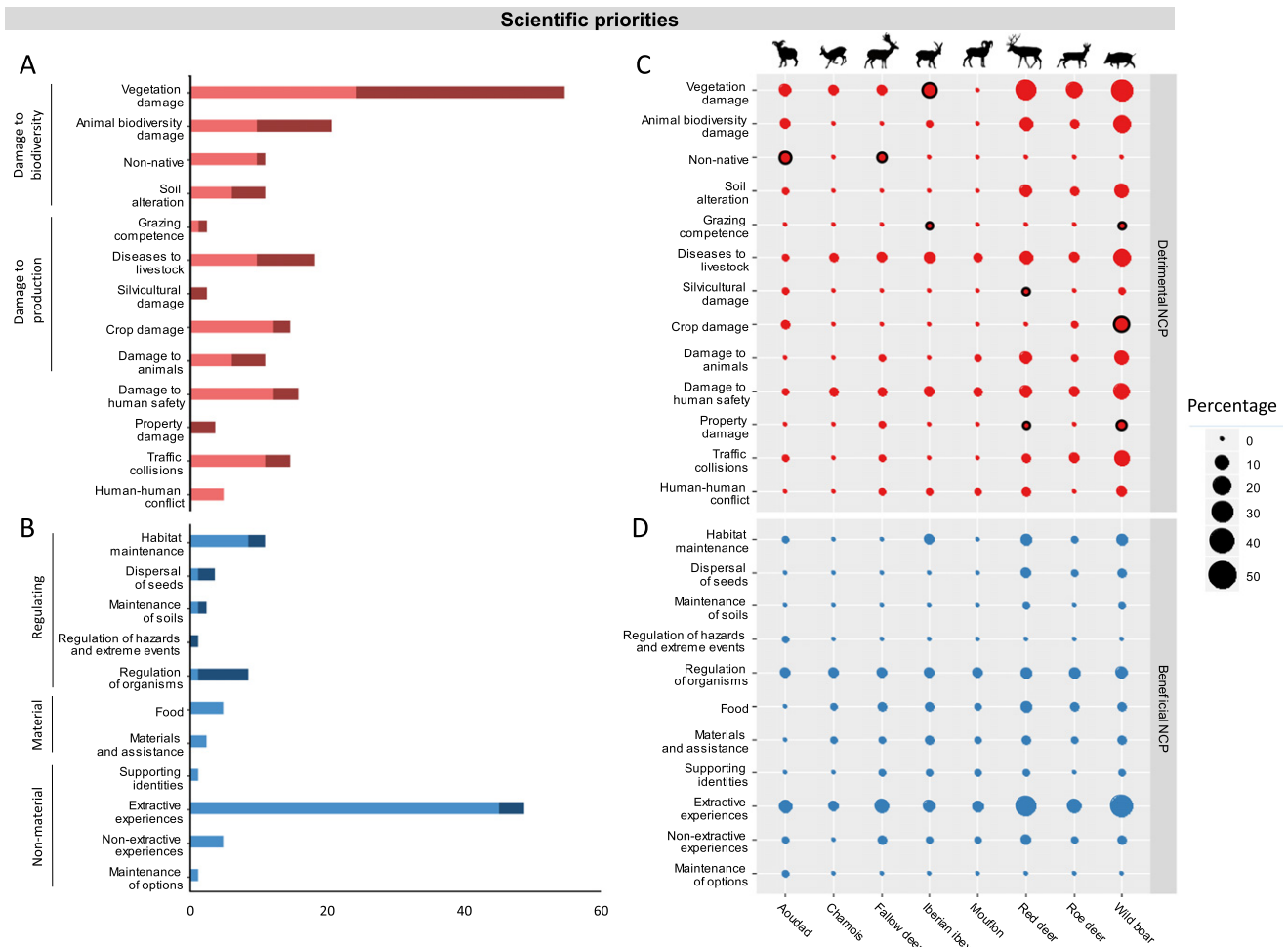


Fig. 3. Percentage of publications that included specific (A) detrimental (red bars; full color: NCP studied; light color: NCP mentioned) and (B) beneficial NCP (blue bars; full color: NCP studied; light color: NCP mentioned). Percentages of publications that included each (C) detrimental (red circles) and (D) beneficial NCP (blue circles) per species. Circles with black borders (C and D) indicate that differences among species were higher than expected ($p < 0.05$) ($n = 82$).

mentioned in the articles: regulation of organisms (8.5%; including ungulate carcasses as important food resource for vertebrate scavengers (6.1%), scavenging by the wild boar (3.7%), ungulates as an alternative prey (2.4%)), habitat maintenance (11.0%), dispersal of seeds (3.7%), maintenance of soils (2.4) and regulation of hazards (1.2%). Material NCP were only mentioned in 4.9% of publications (and never studied), being ungulates conceived as a source of food (4.9%) and as materials (2.4%; Fig. 3B). Table 2 and Fig. 3B present the beneficial NCP provided by ungulates as found in the scientific literature.

We also found associations between particular detrimental NCP and the species causing them ($\chi^2 = 156.6$, $df = 84$, $p < 0.001$). Particularly, we found significant positive association ($p < 0.05$) between vegetation damage and the Iberian ibex and the wild boar ($\chi^2 = 3.2$); non-native species and the aoudad ($\chi^2 = 44.4$) and the fallow deer ($\chi^2 = 14.6$); and crop damage and the wild boar ($\chi^2 = 5.8$). Some of the detrimental NCP less addressed by scientific literature acquired importance with some species, such as the association between grazing competence and both the Iberian ibex ($\chi^2 = 5.7$) and the wild boar ($\chi^2 = 0.1$), silvicultural damage and the red deer ($\chi^2 = 0.2$), and property damage and both the red deer ($\chi^2 = 0.01$) and the wild boar ($\chi^2 = 0.1$). Damage to animal biodiversity, soil alteration, disease transmission to livestock, and damage to animals and to humans, as well as traffic collisions and human-human conflicts were not associated to any species (Fig. 3C). We did not find any association between the species studied and the beneficial NCP ($\chi^2 = 64.1$, $df = 70$, $p = 0.68$; Fig. 3D).

We found no association between the mean number of detrimental NCP provided by wild ungulates included in publications and the distribution of ungulate species in Peninsular Spain ($\rho = 0.29$, $p = 0.49$; Fig. A.4A). However, the mean number of beneficial NCP provided by wild ungulates was negatively correlated with their distribution ($\rho = -0.75$, $p = 0.03$; Fig. A.4B).

3.2. Shepherds' perceptions of NCP provided by wild ungulates

A higher percentage of shepherds mentioned some detrimental NCP caused by wild ungulates (89.1%), whereas 15.2% of the shepherds mentioned at least one beneficial NCP. Among the detrimental NCP identified by shepherds, damage to production was the most mentioned (79.9% of shepherds). Particularly grazing competence (37.5%), damage to animals (37.0%; which was related to the role of wild boar as a predator and it included direct physical damage to livestock (19.0%), damage to game species and their hatchlings and eggs (4.3%), and non-specified species (13.6%)), crop damage (32.1%) and disease transmission to livestock (25.5%). Shepherds also highlighted damage to biodiversity (40.2%), especially vegetation damage (39.7%); and damages to their properties (7.6%; Fig. 4A). The way shepherds identified and described the detrimental NCP provided by wild ungulates is presented in Table 1.

Among beneficial NCP, shepherds mostly mentioned regulating NCP (9.2%), such as regulation of organisms (7.6%; including scavenging by the wild boar (6.0%) and ungulates as an alternative prey (1.6%)),

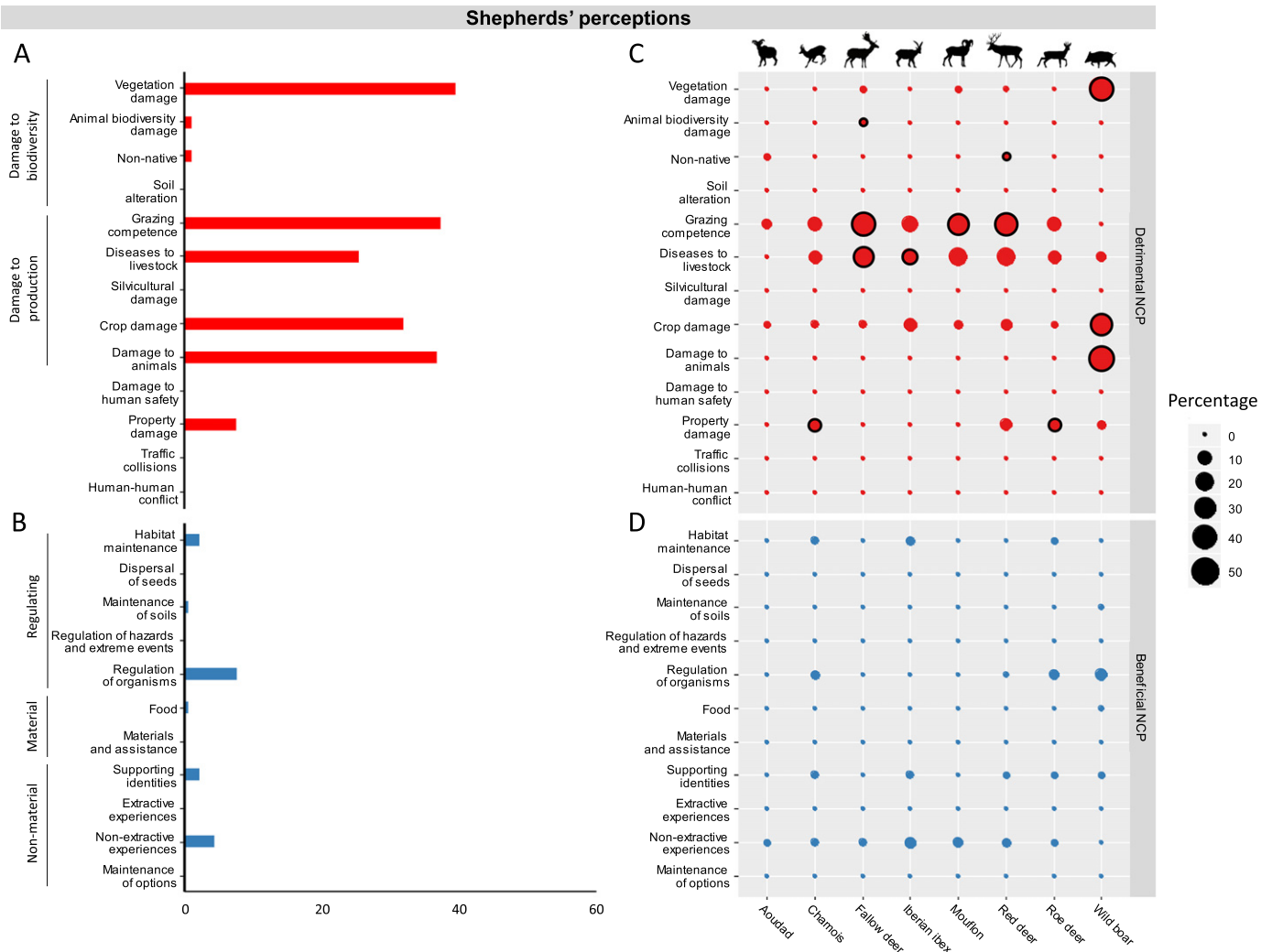


Fig. 4. Percentage of shepherds that perceived specific (A) detrimental (red bars) and (B) beneficial NCP (blue bars). Percentages of shepherds that perceived specific (C) detrimental (red circles) and (D) beneficial NCP (blue circles) per species. Circles with black borders (C and D) indicate that differences among species were higher than expected ($p < 0.05$) ($n = 184$).

habitat maintenance (2.2%) and maintenance of soils (0.5%). Non-material NCP were mentioned by 6.5% of shepherds, particularly non-extractive experiences (i.e. aesthetic value of wild ungulates; 4.3%) and supporting identities (2.2%). Material NCP, particularly meat from ungulates, were mentioned by 0.5% of shepherds (Fig. 4B). Table 2 present the beneficial NCP provided by ungulates as perceived by shepherds.

We found positive associations between several species and the detrimental NCP perceived by shepherds ($\chi^2 = 401.8$, $df = 49$, $p < 0.001$). We found associations ($p < 0.05$) between: rooting behaviour (vegetation damage) and the wild boar ($\chi^2 = 39.7$); grazing competence with livestock and the fallow deer ($\chi^2 = 7.6$), the mouflon ($\chi^2 = 6.2$) and the red deer ($\chi^2 = 5.1$); disease transmission to livestock and both the fallow deer ($\chi^2 = 3.9$) and the Iberian ibex ($\chi^2 = 3.7$); crop damage and the wild boar ($\chi^2 = 11.7$); and damage to animals and the wild boar ($\chi^2 = 65.2$). Some of the less mentioned detrimental NCP acquired relevance for some particular species, such as damages to animal biodiversity and the fallow deer ($\chi^2 = 2.1$); non-native species and the red deer ($\chi^2 = 9.0$); property damage and both the chamois ($\chi^2 = 17.7$) and the roe deer ($\chi^2 = 22.5$; Fig. 4C). We did not find significant associations between particular species and the beneficial NCP perceived by shepherds ($\chi^2 = 33.8$, $df = 35$, $p = 0.53$; Fig. 4D).

We found no association between the mean number of detrimental NCP provided by wild ungulates perceived by shepherds and the distribution of ungulate species in the study areas ($\rho = 0.57$, $p = 0.14$;

Fig. A.4C). However, the mean number of beneficial NCP provided by wild ungulates was positively correlated with their distribution ($\rho = 0.71$, $p = 0.047$; Fig. A.4D).

The average shepherds' perceptions of wild ungulates as providers of NCP (NCP index) varied among species ($\chi^2 = 215.9$, $df = 7$, $p < 0.001$). Whilst the wild boar was assessed by shepherds with the lowest score (NCP index = 1.42), indicating that this species was mostly considered as harmful; the chamois, the roe deer and the Iberian ibex were assessed with the highest scores (NCP index ≥ 3.44),

Table 3

Shepherds' perceptions of wild ungulates as providers of NCP for each species. It shows Nature's contributions to people provider (NCP index), standard deviation (SD), and Tukey posthoc test grouping species according to their NCP index. Different letters indicate significant differences among species (Tukey posthoc test, $p < 0.05$).

	NCP index	SD	Tukey posthoc test
Chamois	3.53	1.16	a
Roe deer	3.51	1.24	a
Iberian ibex	3.44	1.44	a
Aoudad	3.00	1.65	a, b
Red deer	2.49	1.32	b
Mouflon	2.28	1.24	b
Fallow deer	2.17	1.13	b
Wild boar	1.42	0.98	c

indicating that they were mostly perceived as providers of beneficial NCP (Table 3).

NCP index varied according to the diet of the wild ungulate species, since strict herbivores were more perceived by shepherds as providers of beneficial NCP than omnivore species, i.e., wild boars (Mann-Whitney *U* test, $p < 0.001$). Moreover, *NCP index* also varied according to the presence of large carnivores in the study area. We found that wild ungulates in areas with large carnivores were perceived by shepherds as more beneficial than in areas without large carnivores (Mann-Whitney *U* test, $p = 0.03$). We did not find differences in *NCP index* between native and non-native species of wild ungulates (Mann-Whitney *U* test, $p = 0.35$).

3.3. Comparison between scientific priorities and shepherds' perceptions

Scientific priorities about detrimental NCP and shepherds' perceptions of detrimental NCP were not correlated (Spearman's $\rho = 0.18$, $p = 0.56$). In fact, some detrimental NCP, such as soil alteration, silvicultural damage, damage to human safety or traffic collisions were only included in the scientific literature, but were not perceived by shepherds (Fig. A.5A). Likewise, we found no correlation between scientific priorities and shepherds' perceptions of beneficial NCP (Spearman's $\rho = 0.34$, $p = 0.31$). In fact, shepherds did not perceive beneficial NCP that appear in the scientific literature, such as dispersal of seeds, regulation of hazards, provision of materials, extractive experiences and maintenance of options (Fig. A.5B).

4. Discussion

Despite the increasing attention to beneficial NCP in the last years (Díaz et al., 2015), most of the scientific attention regarding ungulates evaluated detrimental NCP (81.7% of articles). This result is consistent with former research on human-wildlife relations that show that publications are biased towards damages and conflicts (Kansky and Knight, 2014; Lozano et al., 2019). It is also remarkable that the few scientific articles addressing beneficial NCP, the majority of research focused on hunting (48.8% of reviewed articles) (e.g. Milner et al., 2006; Schley et al., 2008). The emphasis in the scientific literature on hunting as a beneficial NCP can have a twofold explanation. On the one hand, hunting has been suggested as a management action to limit the populations of particular species that cause any damage (i.e. detrimental NCP). For example, recreational hunting of wild boars has been identified among other measures to control its population across Europe (e.g. Massei et al., 2014). This control tool was also applied to the management of red, fallow and roe deer (e.g. Ramos et al., 2006; Sobalak, 2008). In this sense, hunting goes beyond the recreational experience (i.e. non-material NCP) and it turns into a management recommendation. On the other hand, the focus on hunting can be also explained by the historical bias towards economic valuation in ecosystem service research (e.g. Gómez-Baggethun et al., 2010; Nieto-Romero et al., 2014). In fact, several studies on ungulates assess the economic value of hunting (e.g. Martínez-Jauregui et al., 2014; Popa et al., 2013).

Similar than the scientific literature, shepherds also perceived more detrimental than beneficial NCP. Different factors determine shepherds' perceptions of wild ungulates as providers of detrimental and beneficial NCP, such as diet and the presence of large carnivores in the livestock farming system. Regarding diet, we found that the only omnivorous species (i.e. the wild boar) was worse valued than strict herbivores because their feeding behaviour may provoke different damages to shepherds' livelihoods. For instance, shepherds in other countries considered the wild boar as the main predator of their herds despite there was no evidence of such predation (but scavenging) on livestock (Herrero and Fernández de Luco, 2003). Concerning the presence of large carnivores (i.e., wolves and brown bears), we found that in places where large carnivores are present, shepherds perceived some wild ungulate species as an alternative prey source and their presence is therefore

considered beneficial because it can reduce carnivore attacks on livestock. This phenomenon has been detected also in other European countries. For example, Sidorovich et al. (2003) and Vos (2000) in Belarus and Portugal, respectively, demonstrated that wolf attacks on livestock increase when wild preys are scarce.

Our results showed that the distribution of ungulate species was related with the mean number of beneficial NCP included in scientific publications and perceived by shepherds, but they presented opposite associations. In the case of scientific priorities, wild ungulate species with more restricted distribution were associated with more beneficial NCP. These could be related with the fact that species with reduce distribution or rare species are linked with positive attitudes towards wildlife (e.g. Bandara and Tisdell, 2005; Morales-Reyes et al., 2018). On the other hand, shepherds perceived more beneficial NCP with wider distributions of species. The most beneficial NCP perceived by shepherds was regulation of organisms (e.g. scavenging or prey alternative). This beneficial NCP in particular could be easier detectable by shepherds and developed efficiently by wild ungulates when they present a broader distribution (Morales-Reyes et al., 2018) and, therefore, they are more abundant (Gaston et al., 2000). Accordingly, former research has showed that the shepherds' perceptions of the provision of NCP is dependent on the spatial distribution and abundance of the providing species (García-Alfonso et al., 2019).

Despite that scientific priorities and shepherds' perceptions highlighted more detrimental than beneficial NCP, there are some mismatches. For example, whilst soil alteration, damage to human safety or traffic collisions were considered each one in >10% of publications, they were not perceived by shepherds. By contrast, shepherds mainly considered vegetation damage, grazing competence, damage to animals, crop damage and disease transmission to livestock as important detrimental NCP. Thus, the most important detrimental NCP perceived by shepherds are related with damage to production. As shepherds' livelihoods strongly depend on their livestock, any potential threat is considered relevant. In fact, a meta-analysis developed by Kansky et al., (2014) concludes that the potential of experience damages can underpin negative perceptions towards wildlife. Nevertheless, recognizing and understanding the different NCP (i.e. benefits and detriments) provided by wild ungulates to people allow for a more defensible and exhaustive assessment of the tradeoffs of coexisting with these species (e.g. Ripple et al., 2014; Morales-Reyes et al., 2018). This could favour and promote the social tolerance towards wild ungulate species through different actions. For example, informative talks that target social actors who perceived mainly detrimental NCP might contribute to promote the social tolerance to wild ungulates. In addition, participatory and deliberative processes that involve multiple social actors in decision-making regarding conservation might contribute to design and implement strategies that favour coexistence (e.g. Kansky et al., 2014; Lozano et al., 2019).

Regarding beneficial NCP, whilst hunting was prominent in the scientific literature, shepherds did not conceived it as an important benefit. However, they considered regulation of other organisms (i.e. the capacity of the wild boar to remove carcasses and the role of other species as alternative prey for predators and as carcasses for scavengers) as the most important beneficial NCP, followed by non-extractive experiences such as the aesthetic value derived from some species. These results are consistent with previous research that found that scavenging is positively valued by shepherds in Spain since it contributes to remove their livestock carcasses from the field (Morales-Reyes et al., 2018); and that ungulates are among the most valued organisms because of their beauty (Frynta et al., 2013). The fact that shepherds did not conceive hunting as a beneficial NCP may be because they do not usually access to big game hunting since it is mostly practiced by non-local hunters (Herruzo and Martínez-Jauregui, 2013).

In addition, we found that even when scientific priorities and shepherds' perceptions match in terms of detrimental and beneficial NCP, the species responsible were different. For instance, shepherds did not consider the wild boar as a competitor for graze (grazing competition),

whereas they perceived wild boars as harmful through rooting behavior and damages to vegetation and crops. Further, whilst disease transmission to livestock was related mainly with the wild boar by scientific priorities (but not significant association), shepherds did not consider this potential risk as relevant. Instead, shepherds emphasized their scavenger behaviour and thus they might consider that wild boars reduce the risk of disease transmission via scavenging. Nevertheless, shepherds perceived other ungulates as potential transmitters of diseases -i.e. the Iberian ibex and the fallow deer-. Despite the mouflon and the red deer were considered as potential disease transmitters to livestock by a high percentage of shepherds, there was no significant association between these species and the risk of disease transmission. It has been demonstrated that taxonomically related species are more susceptible to infection by the same parasite species (Freeland, 1983). This factor could be crucial in the wildlife-livestock pathogen interaction as happened with sheep and bighorn (*O. canadensis*) (Wehausen et al., 2011) or between cattle (*Bos taurus*) and American bison (*Bison bison*) (Kilpatrick et al., 2009). Therefore, shepherds' perceptions might be influenced by the taxonomic relation which is reflected in the physical similarity, particularly between the Iberian ibex and the domestic goat (*C. hircus*), both belonging to *Capra* genus, and also with mouflons and sheep (*O. aries*) both belonging to the *Ovis* genus.

Our findings do not only demonstrate that scientific priorities regarding biodiversity and beneficial NCP differ from shepherds interests, needs and perceptions (e.g. Martín-López et al., 2007; Iniesta-Arandia et al., 2014), but also that this mismatch can be determined by the absence of experiencing particular NCP. Furthermore, our results also show the relevance of considering shepherds' local knowledge as their understandings might be different than the scientific knowledge as highlighted by the NCP approach (Díaz et al., 2018). Nevertheless, there is also evidence about the correspondence between local knowledge held by shepherds and scientific knowledge on ecological processes, in particular on the scavenging service provided by scavengers (Morales-Reyes et al., 2019). Former research has also highlighted the usefulness of local knowledge to complement assessments of habitat use by mammals (Prado et al., 2014), or to obtain information on species abundance and population trends (Anadón et al., 2009), whose results are applicable to conservation policies and sustainable management measures. By contrast, previous research has found that shepherds' perceptions regarding specific detrimental NCP (i.e. predation on livestock) do not correspond with the actual ecological processes (i.e. scavenging; Herrero and Fernández de Luco, 2003). For example, Ranglack et al. (2015) quantified the relative grazing impacts of bison, cattle and lagomorphs in an area where ranchers perceived the bison as the main competitor of cattle, but they found that lagomorphs were the largest consumers of biomass. Therefore, our results show evidence of the complementarity of both knowledge-systems. It would be interesting to include the perceptions of other stakeholders not considered in the present study (i.e. farmers, naturalists, managers or conservationist associations) for a more holistic perspective in relation of the positive and negative effects of nature on human well-being. Even so, this study supports the recent calls of IPBES to integrate local knowledge and scientific knowledge in the evaluation of biodiversity and nature's contributions to people (Tengö et al., 2017).

5. Conclusions

The emphasis on detrimental NCP rather than beneficial NCP made by scientists and shepherds can reinforce the idea that ungulates can threaten humans, instead that contribute to our quality of life. This could affect directly the conservation of ungulates in the current rewilding process in settings where stakeholders are involved in the development and implementation of management plans and highlighted only the damages caused by these species. Yet, some discrepancies were found between scientific priorities and shepherds' perceptions regarding the beneficial and detrimental NCP provided by ungulates. For

example, those detrimental NCP mostly stated by shepherds (i.e. grazing competence and damage to animals) were little considered by the scientific literature. The fact that scientific research does not address the interests of shepherds can affect the social tolerance towards wildlife as the damages experienced or perceived by shepherds are not assessed. Therefore, future research on human-ungulate relations should focus on two key aspects: the beneficial NCP provided by ungulates and the existing damages experienced by shepherds but are neglected in the scientific literature. This research agenda will contribute to drift current perceptions about ungulates as sources of detrimental NCP to providers of beneficial NCP.

Declaration of competing interest

The authors declare that there is no conflict of interest.

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Appendix A. Supplementary data

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